Investigating the Impact of REPA Loss on Training Speed and Generation Quality in Diffusion Models

Evgenii Pakhomov
HSE University

Dmitrii Gromak
HSE University

Sergei Kopylov
HSE University

Miron Makhlin
HSE University

Vladimir Novikov
HSE University

Abstract

Diffusion models have emerged as the dominant paradigm for high-fidelity image generation, yet their training remains computationally expensive and slow to converge. Recent work introduced REPA (REPresentation Alignment), a regularization technique that aligns projections of noisy input hidden states in diffusion transformers with clean image representations from pretrained visual encoders. Reported as an ICLR 2025 Oral, REPA demonstrated substantial improvements in training efficiency, achieving up to 17.5x faster convergence on large-scale ImageNet experiments, while also improving final generation quality. In this project, we aim to present a partial reproduction and systematic extension of REPA under constrained computational settings. Rather than scaling to full ImageNet, we plan to reproduce the core REPA framework on reduced datasets and smaller model configurations, verifying its key claim of accelerated convergence relative to standard diffusion training. Building on this, we plan to investigate REPA in low-data regimes, considering both subsampled large-scale datasets (e.g., ImageNet) and naturally small, domain-specific datasets such as Stanford Cars. We will study three main questions: (1) whether REPA's improvements in convergence speed and sample quality persist when training data is limited; (2) how the choice of pretrained feature extractor – comparing alternatives such as DINOv2 and CLIP – affects performance, particularly in narrow visual domains; and (3) how sensitive REPA is to dataset scale, including whether its alignment objective can partially compensate for reduced data availability. Through these experiments, we aim to better characterize the practical benefits and limitations of representation alignment for efficient diffusion training in realistic, resource-constrained settings.

1 Introduction

Diffusion models have become one of the main paradigms for high-quality image generation, with successful architectures ranging from U-Net denoisers to latent diffusion models and diffusion transformers [16, 4, 15, 13]. However, training such models remains computationally expensive: not only must the model learn the denoising dynamics, but it must also be able to extract useful internal visual representations (discriminative features) in its hidden states [8, 20, 1, 11]

Recent work on REPA [21] proposes to accelerate diffusion training by explicitly aligning intermediate hidden states of the denoising model with clean image representations extracted by a frozen pretrained visual encoder. This suggests a simple but powerful idea: instead of forcing the diffusion model to discover good internal representations only through the denoising loss, we can use an external representation model as an auxiliary teacher.

In this project, we study representation alignment as a training principle for efficient diffusion modeling under constrained computational settings. We focus on reduced datasets, low-data regimes, and fine-grained visual domains. At first glance, one might expect REPA to be especially useful in such settings, since the pretrained teacher can inject visual knowledge that is unavailable from the small training set alone. However, this advantage is not obvious. Recent evidence from iREPA [18] suggests that the effectiveness of REPA is determined not merely by the global quality of the teacher encoder, but by the spatial structure of its features. Therefore, in narrow domains, a teacher encoder

Project for the Methods of Optimization in Machine Learning course, HSE University.

may provide globally meaningful but locally mismatched representations, which can make the alignment objective less helpful or even harmful.

Our goal is to characterize when REPA improves convergence speed and sample quality, how its effectiveness depends on the pretrained feature extractor and dataset scale, and under which conditions the alignment objective can become a limitation rather than a reliable source of inductive bias. We also will try to verify the usefulness of REPA loss for text-diffusion models, as recently shown by [6].

2 Related work

Early attempts to bring representation information into diffusion models were mostly indirect. One line of work sought to make diffusion models themselves stronger representation learners, for example by reformulating score matching for unsupervised representation learning, sharing generative and discriminative parameters in joint diffusion models, or imposing bottlenecks and denoising redesigns that encourage semantically meaningful features [10, 2, 5, 1].

Some researchers tried injecting external semantic representations through multi-stage pipelines rather than direct hidden-state matching: Wuerstchen [14] learns a compact semantic image latent that guides diffusion, while Representation-Conditioned Generation [9] first generates self-supervised semantic codes and then conditions image synthesis on them.

REPA differs by making alignment explicit inside a single diffusion model, matching noisy intermediate denoiser features to clean pretrained visual embeddings and using this match as an auxiliary training signal.

iREPA further studies why some teacher encoders are more effective for representation alignment than others. Its main conclusion is that global semantic strength, such as classification accuracy, is not sufficient to explain the benefit of REPA. Instead, spatial structure in patch-level representations appears to be more important for image generation. This is especially relevant for our project, since fine-grained and narrow-domain datasets may require local details that are not well captured by a generic teacher encoder.

Subsequent work has studied how this idea should be adapted across architectures rather than transferred unchanged. In particular, U-REPA [19] shows that moving REPA to diffusion U-Nets is nontrivial because skip connections, stage-specific block roles, and spatial downsampling make transformer-style tokenwise alignment less effective; to address this, it aligns middle-stage U-Net features, upsamples them after projection, and adds a manifold loss to better bridge U-Net and ViT feature spaces.

REPA-E [7] extends the same principle to end-to-end latent diffusion transformers by jointly tuning the VAE tokenizer and the diffusion model, showing that representation alignment can stabilize a setting where diffusion loss alone is ineffective.

More recently, REPA-style alignment has also been adapted to masked diffusion language models [6], suggesting that representation alignment is evolving from a transformer-specific technique into a broader auxiliary training principle across generative architectures and even modalities.

3 Task setting

We follow the notation of REPA. Let $x \sim \mathcal{D}$ be a clean image from the training distribution, and let

$$z = E(x)$$

be its latent representation produced by the encoder E of a pretrained autoencoder. The base diffusion model follows the SiT velocity-prediction formulation used in REPA. Given Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$, we define an interpolation path

$$z_t = \alpha_t z + \sigma_t \epsilon,$$

where $t \in [0, 1]$ and α_t, σ_t determine how the clean latent is gradually transformed into noise. The training target is the velocity of this path with respect to t:

$$u_t(z, \epsilon) = \frac{dz_t}{dt} = \frac{d\alpha_t}{dt} z + \frac{d\sigma_t}{dt} \epsilon.$$

The velocity model $v_\theta(z_t, t)$ is trained to approximate this target:

$$\mathcal{L}_{\text{diff}}(\theta) = \mathbb{E}_{z,\epsilon,t}\left[\|v_\theta(z_t, t) - u_t(z, \epsilon)\|_2^2\right].$$

For the linear interpolation commonly used in this setting, $\alpha_t = 1 - t$ and $\sigma_t = t$, so $u_t(z, \epsilon) = \epsilon - z$.

REPA adds an auxiliary representation-alignment objective. Let f be a frozen pretrained visual encoder, and let

$$y = f(x) \in \mathbb{R}^{N \times C}$$

be the clean image representation extracted by this encoder, where N is the number of patches and C is the representation dimension. Let $H_\theta^\ell(z_t, t)$ denote the intermediate hidden states of the diffusion transformer at layer ℓ. A trainable projection head h_ψ maps these hidden states into the teacher representation space.

The REPA loss aligns projected noisy hidden states with clean teacher features by maximizing patch-wise similarity:

$$\mathcal{L}_{\mathrm{REPA}}(\theta, \psi) = -\mathbb{E}_{x,\epsilon,t}\left[\frac{1}{N}\sum_{n=1}^{N} \mathrm{sim}\left(y_n, h_\psi(H_\theta^\ell(z_t, t))_n\right)\right],$$

where n indexes spatial patches and $\mathrm{sim}(\cdot, \cdot)$ denotes the similarity function used for representation alignment.

The final objective is

$$\mathcal{L}(\theta, \psi) = \mathcal{L}_{\mathrm{diff}}(\theta) + \lambda \mathcal{L}_{\mathrm{REPA}}(\theta, \psi),$$

where λ controls the strength of the alignment term.

Our main experimental question is whether this auxiliary alignment term improves training in small-scale and narrow-domain settings. We will compare the vanilla diffusion objective against REPA-style training under matched architecture, optimizer, data, and compute budgets. We will also vary the frozen representation encoder f and study whether its representations provide useful guidance for generation.

This setting allows us to test two competing hypotheses. The optimistic hypothesis is that REPA should be especially useful in low-data regimes because the pretrained encoder supplies an external representation prior that the diffusion model would otherwise need to learn from limited data. The skeptical hypothesis is that this prior may be mismatched. In particular, a teacher encoder can encode strong global semantic information while still having patch-level representations that are poorly aligned with the spatial structure required in a narrow domain. In that case, minimizing $\mathcal{L}_{\mathrm{REPA}}$ may force the diffusion model to match representations that are globally meaningful but locally suboptimal for generation, reducing or even reversing the expected benefit of REPA.

4 Metrics and Evaluation

We evaluate models along three main axes.

Generation quality. The primary metric for image generation is Fréchet Inception Distance (FID) [3], which measures the distributional gap between generated and real images in the feature space of a pretrained Inception-v3 network. Lower FID indicates better quality and diversity together. We will complement FID with qualitative inspection of random samples, since models with similar FID scores can differ noticeably in perceptual quality – especially on fine-grained domains where structural artifacts may not be captured by Inception features. For Stanford Cars, we will additionally assess part-level coherence, texture realism, and shape consistency. We will also assess diversity, which is important for image generation. For this we will use the mean pairwise LPIPS distance over a set of randomly sampled image pairs from the generator:

$$\mathrm{diversity} \approx \frac{1}{K}\sum_{\substack{i,j=1 \\ i \neq j}}^{K} \mathrm{LPIPS}(x_i, x_j),$$

where higher values indicate greater perceptual spread across samples. We will additionally inspect samples visually for mode collapse.

Training efficiency. Following the evaluation protocol of the original REPA paper [21], we will plot FID as a function of both training steps and wall-clock time, evaluating checkpoints at fixed intervals (e.g., every 20k steps). This will allow us to compare convergence speed independently from final quality: REPA's core promise is that alignment accelerates the early training phase, so a model with REPA should reach a target FID threshold significantly faster than the baseline. We will report both the step count and GPU-hours required to reach representative FID thresholds as scalar efficiency indicators.

Representation behavior. To understand *why* REPA helps or fails in a given setting, we will try to track several internal diagnostics throughout training. For example, we plan to log the alignment loss $\mathcal{L}_{\text{REPA}}$ alongside the diffusion loss $\mathcal{L}_{\text{diff}}$, as well as monitor the mean cosine similarity between projected hidden states $h_\psi(H_\theta^\ell)$ and teacher patch embeddings y. We will also measure linear probing accuracy on the intermediate hidden states of the diffusion model (frozen at various checkpoints) as an indicator of whether the denoiser is learning discriminatively useful representations. Together, these diagnostics help distinguish settings where REPA successfully transfers teacher representations from settings where alignment is achieved superficially without improving generation.

Text diffusion metrics (if time allows). For the masked text diffusion extension, we will use perplexity on a held-out validation set as the primary metric, following [6, 17]. We will also track the alignment loss and cosine similarity between hidden states and teacher token embeddings to verify that the alignment objective is active and meaningful during training.

5 Plan

Step 1: Baseline reproduction. We plan to begin by implementing a self-contained REPA training pipeline on a reduced subset of ImageNet (10% of classes, ≈ 130k images) at 64×64 resolution, using a SiT-S/2 backbone [21] and DINOv2-B [12] as the frozen teacher. We will train two models under identical conditions – one with the REPA loss and one without – for 400k steps and compare their FID-vs-step curves. Success here means reproducing the qualitative trend of faster convergence with REPA, even if the absolute FID numbers differ from the full-scale result. This phase will validate our infrastructure and give us a reliable baseline for subsequent ablations.

Step 2: Low-data and fine-grained domain study. We plan to investigate two complementary low-data regimes. First, we will subsample ImageNet at four data fractions (1%, 5%, 10%, 20%) and train REPA vs. baseline models to the same step budget, measuring how FID and convergence speed scale with dataset size. We hypothesize that REPA's benefit is largest at moderate data regimes (5–20%) where the encoder prior can compensate for limited data, but may saturate at very low fractions where neither model can generate coherent images. Second, we will train on Stanford Cars ($\approx$8k images), a fine-grained benchmark where object geometry and part structure matter more than global semantics. This will test the hypothesis from iREPA [18] that teachers with strong global features but weak spatial structure (e.g., CLIP) may be less helpful – or even harmful – in domains that require precise local detail.

Step 3: Teacher encoder ablation. We plan to compare three pretrained visual encoders as REPA teachers on both the subsampled ImageNet (10%) and Stanford Cars settings: DINOv2-B [12] (strong spatial structure, patch-level features), CLIP ViT-B/32 (strong global semantics, weaker patch structure), and a supervised ResNet-50 (conventional discriminative baseline with no patch-level design). For each teacher, we will measure FID, the alignment loss trajectory, and the linear probing accuracy of the aligned hidden states. This ablation will directly test iREPA's spatial-structure hypothesis in our constrained settings, and determines whether teacher choice should depend on dataset domain.

Step 4: Text diffusion extension. Motivated by [6], who showed that REPA accelerates training of Portuguese Masked Diffusion Language Models (MDLM [17]) by 28.6% in perplexity and confirmed that mid-level alignment with a modern language encoder works best, we plan to adapt this finding to English-language masked diffusion. Concretely, we want to train a small MDLM on a text corpus (e.g., WikiText-103) with and without a REPA-style alignment loss. Following [6], we will align intermediate transformer blocks of the MDLM with token representations from a frozen pretrained encoder. We will compare two teacher choices: a bidirectional encoder (BERT-based) and a decoder-based model (GPT-based), mirroring the BERTimbau vs. Qwen comparison from the original paper. We will evaluate perplexity on the validation set at fixed training step checkpoints to assess whether the convergence speedup observed for Portuguese generalizes to English and to a different model scale.

6 Expected outcomes

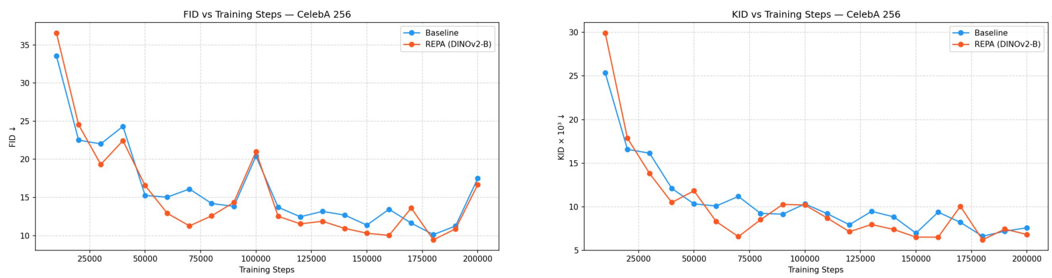
The central output of the project is an empirical characterization of when and why REPA works under constrained settings. For the image experiments, we expect REPA's convergence benefit to be robust at moderate data scales (5–20% of ImageNet) but to degrade in very low-data regimes, where the alignment signal becomes too weak relative to the denoising objective to provide a reliable training prior. On fine-grained domains such as Stanford Cars, we expect teacher choice to matter substantially: teachers with strong spatial patch structure (DINOv2) should outperform those with primarily global semantic features (CLIP), consistent with the findings of iREPA [18]. More broadly, we may find that a teacher which ignores domain-specific visual features can actively harm training by pulling hidden states toward representations that are globally coherent but locally mismatched to the target distribution.

For the text diffusion extension, we expect the qualitative finding of [6] to generalize beyond Portuguese: mid-level alignment with a bidirectional encoder should reduce perplexity and accelerate convergence compared to the unaligned baseline, while decoder-based teachers may prove less effective due to their weaker bidirectional token structure. A negative or null result would itself be informative, suggesting that the REPA benefit in text diffusion is tied to specific linguistic or resource conditions rather than being a general property of representation alignment.

Taken together, the project should clarify whether REPA is best understood as a universal training accelerator or as a representation prior whose practical benefit depends jointly on the teacher's feature structure, the scale and domain of the training data, and the alignment design choices. If REPA proves fragile across these axes, this would motivate follow-up work on adaptive or domain-aware alignment strategies.

(a) Imagenette (b) CelebA

Figure 1: Memorization examples on Imagenette and CelebA. See full examples in Appendix A.



(a) FID (b) KID

Figure 2: FID and KID plots for training checkpoints

7 Baseline Reproduction

We begin by reproducing the key results of the original REPA paper: faster convergence in terms of FID and improved stability of model features during training.

7.1 Imagenette

We first train on Imagenette – a subset of ImageNet containing 10 easily classified classes with approximately 1 000 samples per class. After 100 000 iterations, the model achieves good generation quality, but at a cost: the model is sufficiently large to memorize the entire dataset, a tendency amplified by the REPA loss, as shown in Figure 1a. This reveals a key risk of applying diffusion models to small-scale datasets – complete training set memorization rather than genuine generalization.

We address this problem by [MIRON].

7.2 CelebA

Since training on small, multi-domain datasets leads to memorization, we investigate whether a single-domain dataset can mitigate this issue. We select the Large-scale CelebFaces Attributes (CelebA) dataset as our fine-grained benchmark and train the diffusion model on it. Notably, the memorization problem does not arise here, as illustrated in Figure 1b, enabling analysis analogous to that in the original REPA paper.

7.2.1 Training Speedup Analysis

Figure 2 shows FID and KID as a function of training iteration. Both models improve steadily in generation quality. Early in training, the REPA model exhibits higher total loss, consistent with the original paper's observation that the alignment term adds overhead while the model is still learning to extract meaningful features. By iteration 30 000 this gap closes, and by iteration 70 000 the REPA model reaches FID ≈ 11 – a threshold the baseline only reaches at iteration 180 000. This corresponds to a $2.5\times$ speedup in early training, corroborated by the KID curve, where the REPA model also attains its all-time minimum at iteration 70 000.

Beyond this point, both models' metrics degrade at a similar rate. Once $\mathcal{L}_{\text{REPA}}$ approaches zero around iteration 70 000, the remaining gradient signal comes entirely from the diffusion loss, causing the two training curves to converge. The periodic quality dips visible in both curves likely reflect transient instabilities from a learning rate that is too high at later stages. It is also worth noting that these metrics are evaluated on the non-EMA model, which amplifies measurement noise.

We attribute the early saturation of REPA's benefit to the relative simplicity of CelebA's spatial structure: facial images share consistent geometry, pose, and composition, allowing the model to absorb the essential representational signal from the teacher within 70 000 iterations, after which training reduces to standard diffusion fine-tuning.

7.2.2 DIMA, feature maps analysis, Appendix B

7.2.3 VOVA, attribute direction vectors analysis, appendix C

References

[1] Xinlei Chen, Zhuang Liu, Saining Xie, and Kaiming He. "Deconstructing Denoising Diffusion Models for Self-Supervised Learning". In: *arXiv preprint arXiv:2401.14404* (2024). URL: https://arxiv.org/abs/2401.14404.

[2] Kamil Deja, Tomasz Trzcinski, and Jakub M. Tomczak. "Learning Data Representations with Joint Diffusion Models". In: *Machine Learning and Knowledge Discovery in Databases: Research Track*. Springer, 2023, pp. 543–559. DOI: 10.1007/978-3-031-43415-0_32. URL: https://doi.org/10.1007/978-3-031-43415-0_32.

[3] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. "GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium". In: *Advances in Neural Information Processing Systems*. Vol. 30. 2017. URL: https://proceedings.neurips.cc/paper/2017/hash/8a1d694707eb0fefe65871369074926d-Abstract.html.

[4] Jonathan Ho, Ajay Jain, and Pieter Abbeel. "Denoising Diffusion Probabilistic Models". In: *Advances in Neural Information Processing Systems*. Vol. 33. 2020. URL: https://proceedings.neurips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html.

[5] Drew A. Hudson, Daniel Zoran, Mateusz Malinowski, Andrew K. Lampinen, Andrew Jaegle, James L. McClelland, Loic Matthey, Felix Hill, and Alexander Lerchner. "SODA: Bottleneck Diffusion Models for Representation Learning". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. June 2024, pp. 23115–23127.

[6] Adalberto Ferreira Barbosa Junior, Lucas Lima Neves, and Adriano César Santana. "Accelerating Portuguese Masked Diffusion Models through Representation Alignment". In: *Proceedings of the 17th International Conference on Computational Processing of Portuguese (PROPOR 2026) - Vol. 1*. Salvador, Brazil: Association for Computational Linguistics, Apr. 2026, pp. 968–973. URL: https://aclanthology.org/2026.propor-1.97/.

[7] Xingjian Leng, Jaskirat Singh, Yunzhong Hou, Zhenchang Xing, Saining Xie, and Liang Zheng. "REPA-E: Unlocking VAE for End-to-End Tuning of Latent Diffusion Transformers". In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. Oct. 2025, pp. 18262–18272.

[8] Daiqing Li, Huan Ling, Amlan Kar, David Acuna, Seung Wook Kim, Karsten Kreis, Antonio Torralba, and Sanja Fidler. "DreamTeacher: Pretraining Image Backbones with Deep Generative Models". In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. Oct. 2023, pp. 16698–16708.

[9] Tianhong Li, Dina Katabi, and Kaiming He. "Return of Unconditional Generation: A Self-supervised Representation Generation Method". In: *Advances in Neural Information Processing Systems*. 2024. URL: https://openreview.net/forum?id=clTa4JFBML.

[10] Sarthak Mittal, Korbinian Abstreiter, Stefan Bauer, Bernhard Schölkopf, and Arash Mehrjou. *Diffusion-Based Representation Learning*. 2024. arXiv: 2105.14257 [cs.LG]. URL: https://arxiv.org/abs/2105.14257.

[11] Soumik Mukhopadhyay, Matthew Gwilliam, Vatsal Agarwal, Namitha Padmanabhan, Archana Swaminathan, Srinidhi Hegde, Tianyi Zhou, and Abhinav Shrivastava. *Diffusion Models Beat GANs on Image Classification*. 2023. arXiv: 2307.08702 [cs.CV]. URL: https://arxiv.org/abs/2307.08702.

[12] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaa El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Herve Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bourdoukos. "DINOv2: Learning Robust Visual Features without Supervision". In: *Transactions on Machine Learning Research* (2024). URL: https://openreview.net/forum?id=a68SUt6zFt.

[13] William Peebles and Saining Xie. "Scalable Diffusion Models with Transformers". In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2023. URL: https://arxiv.org/abs/2212.09748.

[14] Pablo Pernias, Dominic Rampas, Mats Leon Richter, Christopher Pal, and Marc Aubreville. "Würstchen: An Efficient Architecture for Large-Scale Text-to-Image Diffusion Models". In: *International Conference on Learning Representations*. 2024. URL: https://openreview.net/forum?id=gU58d5QeGv.

[15] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. "High-Resolution Image Synthesis with Latent Diffusion Models". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2022, pp. 10674–10685.

[16] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. "U-Net: Convolutional Networks for Biomedical Image Segmentation". In: *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*. Springer, 2015, pp. 234–241. DOI: 10.1007/978-3-319-24574-4_28.

[17] Subham Sekhar Sahoo, Marianne Arriola, Aaron Gokaslan, Edgar Mariano Marroquin, Alexander M. Rush, Yair Schiff, Justin T. Chiu, and Volodymyr Kuleshov. "Simple and Effective Masked Diffusion Language Models". In: *Advances in Neural Information Processing Systems*. 2024. URL: https://openreview.net/forum?id=L4uaAR4ArM.

[18] Jaskirat Singh, Xingjian Leng, Zongze Wu, Liang Zheng, Richard Zhang, Eli Shechtman, and Saining Xie. "What Matters for Representation Alignment: Global Information or Spatial Structure?" In: *International Conference on Learning Representations*. 2026. URL: https://openreview.net/forum?id=y0UxFtXqXf.

[19] Yuchuan Tian, Hanting Chen, Mengyu Zheng, Yuchen Liang, Chao Xu, and Yunhe Wang. *U-REPA: Aligning Diffusion U-Nets to ViTs*. 2026. arXiv: 2503.18414 [cs.CV]. URL: https://arxiv.org/abs/2503.18414.

[20] Weilai Xiang, Hongyu Yang, Di Huang, and Yunhong Wang. "Denoising Diffusion Autoencoders are Unified Self-supervised Learners". In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 2023, pp. 15756–15766.

[21] Sihyun Yu, Sangkyung Kwak, Huiwon Jang, Jongheon Jeong, Jonathan Huang, Jinwoo Shin, and Saining Xie. "Representation Alignment for Generation: Training Diffusion Transformers Is Easier Than You Think". In: *International Conference on Learning Representations*. 2025. URL: https://openreview.net/forum?id=DJSZGGZYVi.